DAY 3 · SATURDAY 24 OCTOBER 2026 · MFB 545

Digital credit, neobanks and financial inclusion

A loan granted in thirty seconds in an app rests on a model that turns data into a probability of default, and on a threshold someone chose. A neobank with fifty million customers can earn more than a billion; another, with more revenue per customer, loses money. This day opens both black boxes: you build a score on 1,000 real loans, set the threshold, redo a neobank's accounts, lend to strangers on a simulated platform, debate buy now, pay later — then design your app's pension-savings plan.

🕘 **09:00–12:15 · 13:30–16:45 · four 90-min blocks** | 💻 **laptop · a Google account for Colab**

🎤 **the talk is short —** 🛠️ **most of the time, you are the ones working**

BY THE END OF DAY 3, YOU WILL BE ABLE TO

- Explain how a credit score turns data into a probability of default, and read an AUC and a Gini.
- Choose an acceptance threshold by trading off default rate, volume and result — and calculate an expected loss.
- Calculate a neobank's ARPU, customer lifetime value and acquisition-cost payback period.
- Bring a neobank's published income statement down to one customer, and say why some lose money.
- Tell the forms of crowdfunding apart, measure the risk of a peer-to-peer loan portfolio, and argue the status of buy now, pay later.
- Design your app's pension-savings plan: payments, fees, capital at 65, transparency and safeguards.
- Explain, having done it, why a shared ledger holds without a keeper: keys, hashes, mining, consensus — and what it does not prove.

BEFORE YOU START — TICK

- [ ] **Signed in** on this site (the button at the bottom right).
- [ ] **Signed in to Google** in the same browser: the Block 1 notebook opens in Google Colab ↗. Without an account, the simulator in section 3.2 is enough.
- [ ] **Your work from Days 1 and 2** at hand: your team's problem, customer, payment flow and revenue model (Day 2, section 2.9).

BLOCK 1 · 09H00–10H30 how a score works · the notebook · threshold versus profitability

3.1

How a credit score works

Expected loss · from data to decision · AUC and Gini · the threshold · KKG, Directive 2023/2225, AI Act, usury rate

🎤 TALK · 20 MIN

3.2

Live scoring · threshold versus profitability

The Colab notebook on 1,000 loans · 300 test loans and the threshold · one applicant and their two reasons

🛠 WORKSHOP · 55 MIN

BLOCK 2 · 10H45–12H15 neobank economics · CAC, LTV, churn · the quiz

3.3

The economics of a neobank

Five revenue lines · ARPU, LTV, CAC, churn · Revolut, N26, neon, Yuh

🎤 TALK · 15 MIN

3.4

CAC, LTV · the income statement per customer

The unit-economics simulator · rebuilding published accounts

🛠 WORKSHOP · 50 MIN

3.5

End-of-morning quiz

Eight questions · launched in class · counts toward participation

✅ QUIZ · 10 MIN

BLOCK 3 · 13H30–15H00 crowdfunding, peer-to-peer lending · the BNPL debate

3.6

Crowdfunding, peer-to-peer lending, BNPL

Four forms · a P2P loan step by step · buy now pay later and its status

🎤 TALK · 20 MIN

3.7

A campaign, a portfolio, a debate

What is left of a campaign · 2,000 simulated years of loans · “is BNPL credit?”

🛠 WORKSHOP · 55 MIN

BLOCK 4 · 15H15–16H45 digital credit in Africa · the pension plan · build your blockchain

3.8

Digital credit in Africa

M-Shwari and what we know about it · M-Baobab · the abuses and Kenya’s response

🎤 TALK · 15 MIN

3.9

Your app’s pension-savings plan

The plan at 65, fees, transparency, safeguards · peer review

🛠 WORKSHOP · 75 MIN

3.10

Build your blockchain

Every student is a node · keys, hashes, mining, payments, forks, cheating · what the ledger proves

🛠 WORKSHOP · 60 MIN

← Course page

reset my Day 3 progress on this device

SECTION 3.1 · BLOCK 1

🎤 TALK · 20 MIN

How a credit score works

Instructor's talk. If it does not take place, read this section on your own, then do the sorting exercise at the bottom of the page.

THE LENDER'S QUESTION

How much will this loan cost me, on average?

Expected loss (EL) = PD × LGD × EAD

- **PD** — *probability of default*: the probability that the borrower stops paying (usually: more than 90 days late) over the life of the loan or over one year. This is what the score estimates.
- **LGD** — *loss given default*: the share of the amount the lender does not get back, after collection. For unsecured consumer credit, often 40 to 70%.
- **EAD** — *exposure at default*: the amount owed at the moment of default.

A CHF 5,000 loan, PD 12%, LGD 50%: expected loss 0.12 × 0.50 × 5,000 = **CHF 300**. The price of credit must cover this loss, the cost of money, servicing costs and the capital tied up. This is **risk-based pricing**: two customers of the same product can pay two different rates.

THE MECHANISM · FROM DATA TO DECISION

Seven steps between the application and the app's "yes"

▸ AN APPLICATION FOR A CHF 3,000 LOAN OVER 24 MONTHS

1 · The application. The customer fills in the app: amount, term, income, expenses, job. Identity is checked by the Day 1 eKYC chain.

2 · External data, with consent. The credit register (ZEK and IKO in Switzerland, a credit information bureau in WAEMU) gives current loans and arrears. If the customer allows it, an **open-banking API** (Day 2, an AISP) reads twelve months of statements; in Africa, the mobile-wallet history plays that role.

3 · The variables. Raw data is turned into a few dozen variables: “negative average balance”, “number of payments more than 30 days late”, “salary paid in regularly”, “expenses-to-income ratio”. This is the step that makes a score good or bad.

4 · The model. A **logistic regression** adds up the weighted variables — the *logit* — then turns it into a probability: PD = $1 \div (1 + e^{-\text{logit}})$. Banks often translate it into **points** (a scorecard where, for example, +20 points double the good-payer odds). Digital lenders also use boosted decision trees, more accurate and harder to explain.

5 · Rules before the score. Some refusals never reach the model: minimum age, residence, detected fraud, an IKO entry, insufficient credit capacity (in Switzerland, the KKG requires that the credit can be repaid within 36 months without touching the subsistence minimum).

6 · Threshold and price. PD below the threshold: accepted, and the rate is set according to the PD (risk-based pricing), within the legal cap. Above: refused — or a smaller amount.

7 · Afterwards. The loan is monitored: every late payment is reported to the register, every repayment feeds the next score. The model itself is revalidated every year on recent loans, because the economy changes and the score degrades.

✓ All 7 steps are shown

MEASURING A SCORE

AUC and Gini: does the score rank the good before the bad?

A score does not say who will default; it *ranks*. It is judged on loans it has never seen. The **AUC** is the probability that a randomly drawn bad payer gets a higher PD than a randomly drawn good payer: **0.5** = a coin toss, **1** = perfect ranking. Banks rather report the **Gini = 2 × AUC – 1**. A consumer-credit score typically has a Gini between 0.5 and 0.7; above 0.9 on test data, suspect a mistake (a variable that “knows” the answer).

The threshold. The model gives a PD; a person sets the *threshold* above which applications are refused. If a repaid loan earns a margin m (as a % of the amount) and a default costs the LGD, a loan is worth making as long as $(1 - \text{PD}) \times m > \text{PD} \times \text{LGD}$, that is:

Theoretical threshold: PD < m ÷ (m + LGD)

With a 15% margin and a 50% LGD: 0.15 ÷ 0.65 = **23.1%**. Accepting more widely raises volume and defaults; accepting more tightly turns away good customers. The simulator in the next section shows the trade-off on real loans.

THE FRAMEWORK

What the law requires of a score, in Switzerland, the EU and WAEMU

- **Switzerland.** The Consumer Credit Act (KKG/LCC) requires a **creditworthiness assessment** and a check with the Consumer Credit Information Centre (**IKO**); the **ZEK** is the lenders' private credit register. Maximum annual effective rate in 2026: **10%** for a cash loan, **12%** for an overdraft or a credit card.
- **European Union.** The new Consumer Credit Directive (2023/2225, applicable from **20 November 2026**) requires a creditworthiness assessment based on relevant data, forbids using sensitive data (health, origin…) and social-media data in it, and gives a customer refused by an automated decision the right to an **explanation** and to **human intervention**. The **AI Act** classes credit scoring of individuals as a **high-risk** AI system. And the Court of Justice ruled in 2023 (SCHUFA, C-634/21) that the score itself can be an automated decision under the GDPR when the lender relies on it decisively.
- **WAEMU.** The **usury rate** set by the WAMU Council of Ministers caps the cost of credit: **14%** a year for banks since 1 June 2026 (15% before), **24%** for microfinance institutions and financial institutions. Credit information bureaus (BIC) licensed by the BCEAO share repayment histories.

CHECK · 4 MIN

Can this data go into a score?

▸ USABLE · SENSITIVE, MUST BE JUSTIFIED · EXCLUDE

Late payments over the last two years, from the credit register

Usable | Sensitive, must be justified | Exclude

The applicant's sex

Usable | Sensitive, must be justified | Exclude

Age

Usable | Sensitive, must be justified | Exclude

Twelve months of bank statements read through an open-banking API, with the customer’s consent

Usable | Sensitive, must be justified | Exclude

The home postcode

Usable | Sensitive, must be justified | Exclude

The phone’s contact list and social-media activity

Usable | Sensitive, must be justified | Exclude

The history of deposits, transfers and payments in a mobile-money wallet, with consent

Usable | Sensitive, must be justified | Exclude

Nationality

Usable | Sensitive, must be justified | Exclude

0 of 8 sorted

SECTION 3.2 · BLOCK 1

🛠 WORKSHOP · 55 MIN

Live scoring · threshold versus profitability

A · 25 MIN · THE NOTEBOOK

Build the score on 1,000 real loans

The public “German Credit” dataset (UCI, 1,000 consumer loans, 300 of which went bad) is the classic training set for scoring. The notebook loads it, looks at the default rate by category, fits a logistic regression on seven variables, measures the AUC and computes the portfolio result threshold by threshold. Run the cells in order (▶ or `Shift + Enter`); there is no line to write.

Open the notebook in Colab ↗ | Download the .ipynb

If Colab asks for confirmation (“This notebook was not authored by Google”), choose “Run anyway”. Without a Google account: do part B, which uses exactly the same model and the same 300 test loans.

Cell 5 of the notebook: what is the **AUC on the test sample**, rounded to two decimals?

… Check

With a 15% margin and a 50% LGD, what is the **theoretical threshold**, as a % PD?

… % Check

✍ **The variable with the highest factor on the odds, and what it means in one sentence. What you saw in cell 7 (sex, nationality), and whether you would use those variables.**

e.g. A poor credit history multiplies the odds of default by 3.1… ; foreign workers have a higher default rate here, but…

B · 20 MIN · THE “THRESHOLD VERSUS PROFITABILITY” SIMULATOR

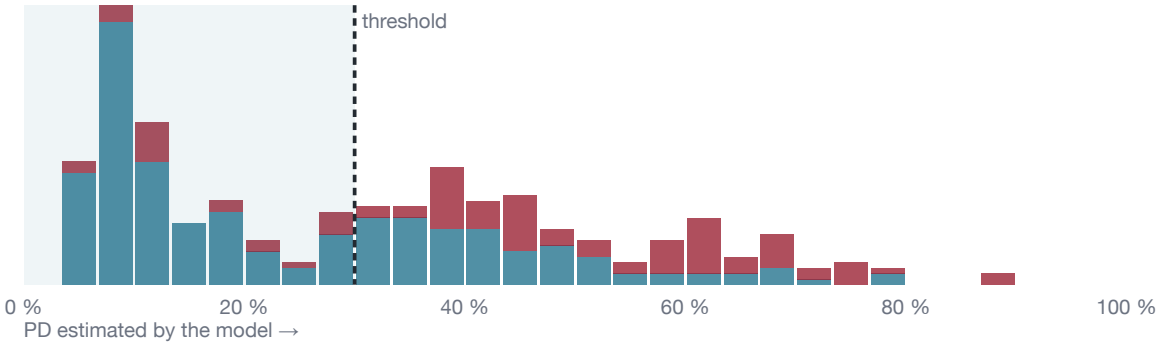
The 300 test loans, and the threshold in your hands

Each bar counts some of the 300 loans the model has not seen, placed by their PD: **blue**, repaid; **red**, defaulted. Everything to the left of the threshold is accepted. Amounts from the dataset, read as francs for the exercise.

Threshold: accept if the PD is below 30 %

Margin on a repaid loan 15 %

Loss given default (LGD) 50 %



152 / 300	13.8 %	76	72 / 93
ACCEPTED	DEFAULTS AMONG ACCEPTED	GOOD CUSTOMERS REFUSED	DEFAULTS AVOIDED

15,831
PORTFOLIO RESULT

Find the best threshold | Copy my choice into my notebook ↓

Accept everyone: 300 loans, 93 defaults (31%), result -99,825. Theoretical threshold with these assumptions: 23.1 %.

Set the margin back to 15% and the LGD to 50%, and put the threshold at **20%**. How many loans out of 300 are accepted?

… loans Check

C · 10 MIN · ONE APPLICANT, ONE DECISION, TWO REASONS

The same model, applied to one person

The notebook's model, with its coefficients. Change the profile: the PD, the decision at the threshold chosen in B, and the two reasons that weigh most. This is what a bank must be

able to tell a refused customer.

Current account

negative balance

Term 24 months

Amount 3,000

Savings below 100

yes

Age 30 years

Poor credit history

no

49.4 %	488	refused
ESTIMATED PD	SCORE IN POINTS	AT THE 30 % THRESHOLD (SECTION B)

The two reasons that raise the risk most, compared with the average applicant: the current account (negative balance) (+0.80 on the logit) ; savings below 100 (+0.18 on the logit). Score in points: 600 for odds of 50 good to 1 bad, +20 points each time the odds double.

An applicant: current account with a negative balance, 24 months, 3,000, savings below 100, aged 30, normal history. What is their **PD**, in %?

… % Check

✍️ **Your threshold, what it gives (accepted, defaults, result), and what you would say to a customer refused at a 30% PD.**

The simulator's button inserts the threshold and its figures; add your reasoning.

DEBRIEF · 15 MIN

Three questions: which threshold did you choose, and what moved it — the margin, the LGD, fear of defaults? How many good customers does the best threshold turn away? Is the score "fair" if it refuses more often a group that defaults more often?

SECTION 3.3 · BLOCK 2

🎤 TALK · 15 MIN

The economics of a neobank

Instructor's talk, or self-study reading.

WHERE THE MONEY COMES FROM

Five revenue lines, all small per customer

- **Interchange** (Day 2): the neobank is the issuer; it earns a share of every card payment. In the EU, capped at 0.2% on debit: CHF 800 of monthly spending brings in CHF 1.60.
- **FX**: a margin on the rate, often zero on weekdays up to a cap, then marked up (at weekends, above the limit, on illiquid currencies).
- **Subscriptions**: a few percent of customers pay for a plan (travel insurance, more free FX, a metal card).
- **Interest margin**: deposits placed with the central bank or in government securities earn interest; the neobank passes part of it on. With rates at zero (Switzerland 2015–2022), this line disappears.
- **Credit and investing**: overdrafts, loans, buy now pay later, brokerage. The most profitable lines — and the ones that need a banking licence and a score.

THE MECHANISM · UNIT ECONOMICS

What a customer is worth, and what they cost to acquire

ARPU = average revenue per active customer per month

Customer lifetime value (LTV) = ARPU × contribution margin ÷ monthly churn

Payback period = CAC ÷ (ARPU × contribution margin)

The **contribution margin** is what remains of ARPU after variable costs (card processing, support, fraud, compliance per customer). **Churn** is the share of customers who leave — or go inactive — each month: at 2% a month, a customer stays 1 ÷ 0.02 = 50 months on average. **CAC** is the cost of acquiring a customer (advertising, referral bonuses, onboarding

and identity-verification costs). The industry rule: **LTV ≥ 3 × CAC**, and CAC recovered in under 12 to 18 months.

FOUR CASES

Revolut, N26, neon, Yuh: why some lose money with millions of customers

- **Revolut** (London, 2015): 52.5 million customers at the end of 2024, revenue £3.1 billion, profit before tax £1.1 billion — a fourth profitable year. The engine: many products per customer (FX, subscriptions, credit, trading, interest) and global scale on a single platform.
- **N26** (Berlin, 2013): 4.2 million "revenue-relevant" customers at the end of 2023, revenue €313.5 million, net loss €102.4 million. Growth held back from 2021 to 2024 by BaFin, which had capped new account openings over anti-money-laundering weaknesses.
- **neon** (Zurich, 2019): about 250,000 customers at the end of 2025; no banking licence of its own (accounts are held by Hypothekarbank Lenzburg), a low-fee, subscription model.
- **Yuh** (2021, Swissquote and PostFinance; 100% Swissquote since July 2025): profitable at the end of 2024, just under 400,000 accounts at the end of 2025. Revenue comes largely from investing and the interest margin rather than from cards.

Sources: the companies' annual reports and press releases (Revolut, Annual Report 2024; N26, press release of 13 Nov. 2024; Swissquote, Annual Report 2025; PostFinance, "Yuh 2024" release). The figures change every year: the workshop asks you to look up the latest published ones.

CHECK · 3 MIN

Where does the neobank's revenue come from, in each case?

▸ INTERCHANGE · FX · SUBSCRIPTION · INTEREST MARGIN · CREDIT

You pay CHF 4.80 for a coffee in Lausanne with your neobank card

Interchange | FX | Subscription | Interest margin | Credit

CHF 2,000 stays in your account all month

Interchange FX Subscription Interest margin Credit

You switch to the Premium plan at 9.99 a month for the travel insurance

Interchange FX Subscription Interest margin Credit

On a Saturday, you change CHF 500 into CFA francs in the app

Interchange FX Subscription Interest margin Credit

You spread the purchase of a laptop over six months in the app

Interchange FX Subscription Interest margin Credit

You pay €120 in a restaurant in Paris with the same card

Interchange FX Subscription Interest margin Credit

0 of 6 sorted

SECTION 3.4 · BLOCK 2

🛠 WORKSHOP · 50 MIN

CAC, LTV, churn · a neobank's income statement

A · 25 MIN · THE UNIT-ECONOMICS SIMULATOR

Build ARPU line by line, then see whether the customer is worth what they cost

A fictitious neobank. Start from one of the three profiles, then move the sliders. Find **the slider that pushes LTV below CAC** the fastest.

EU neobank, card only | Swiss neobank | Multi-product

Card spending per month 800 CHF

Interchange 0.20 %

Share of customers on a paid plan 10 %

Plan price per month 9.00 CHF

Average deposit 2,000 CHF

Annual interest margin on deposits 1.50 %

FX per month 150 CHF

FX margin 0.50 %

Variable cost per customer per month 2.50 CHF

Monthly churn 1.50 %

Acquisition cost (CAC) 80 CHF

PER ACTIVE CUSTOMER PER MONTH	CHF	SHARE OF ARPU
Interchange	1.60	28 %
Subscriptions	0.90	16 %
Interest margin	2.50	43 %
FX	0.75	13 %
ARPU	**5.75**	**100 %**

– variable cost	-2.50	
Contribution	**3.25**	

67 months	217	2.71	24.6 months
AVERAGE LIFETIME	LTV (CHF)	LTV ÷ CAC (TARGET ≥ 3)	PAYBACK PERIOD

Illustrative profiles, built to show the mechanisms: these are not the figures of any real bank.

A neobank: ARPU CHF 6 a month, contribution margin 60%, churn 2% a month. What is the **LTV**?

… CHF Check

The same, but churn rises to **4%** a month (customers open the account for the bonus, then abandon it). LTV?

… CHF Check

With a CAC of CHF 90 and the same contribution of CHF 3.60 a month, what is the **payback period**?

… months Check

✍ **Which slider pushes LTV below CAC the fastest? What would a neobank actually do to fix it?**

e.g. Churn: from 1.5% to 3% a month, LTV is halved… ; the fix: make the account the main account (salary paid in)…

B · 25 MIN · IN PAIRS · THE INCOME STATEMENT PER CUSTOMER

Rebuild a neobank's accounts from its published figures

Two columns are prefilled with published figures. In the third, put a neobank of your choice — neon, Yuh, Wise, Monzo, Nubank, or an African neobank (Kuda, TymeBank,

Moniepoint…) — using **its latest annual report or press release**, found online now. The table brings everything down to one customer.

	REVOLUT · 2024	N26 · 2023	your neobank · year
Currency	£	€	CHF
Customers (millions)	52.5	4.2 "revenue-relevant"	
Revenue (millions)	3,100	313.5	
Result (millions)	1,100 before tax	–102.4 net	
Revenue per customer	**£ 59.05**	**€ 74.64**	
Costs per customer (revenue – result)	**£ 38.10**	**€ 99.02**	
Result per customer	**£ 20.95**	**€ -24.38**	
Source	Annual Report 2024	press release, 13.11.2024	link or document

Copy the table into my notebook ↓

Revolut 2024: what is the **revenue per customer**, in pounds?

… £ Check

N26 2023: what is the **net result per customer**, in euros?

… € Check

✍️ **The table, your source, then: N26 has more revenue per customer than Revolut and loses money. Why? And your neobank?**

SECTION 3.5 · BLOCK 2

✅ QUIZ · 10 MIN

End-of-morning quiz

INTERACTIVE QUIZ · COUNTS TOWARD PARTICIPATION

Eight questions, one attempt each, all explained

Launched in class; this page unlocks by itself. On your own, without an AI assistant. A missed quiz scores zero unless arranged in advance.

🔒 **The quiz starts together.** It is launched in class; this page unlocks by itself within seconds, without reloading.

0 / 8 answered

1. **A CHF 10,000 loan, PD 5%, LGD 60%. The expected loss is…**

- CHF 500
- CHF 6,000
- CHF 300
- CHF 30

2. **A credit score has an AUC of 0.5 on the test sample. This means…**

- it ranks no better than chance
- it is wrong about defaults half the time
- it accepts 50% of applications
- it is excellent: 0.5 is the norm

3. **The lender raises its acceptance threshold (it accepts higher PDs). In general…**

volume falls and the default rate falls

nothing changes if the model is good

volume falls and the default rate rises

volume rises and the default rate among accepted loans rises

4. A 15% margin on a repaid loan, a 50% LGD. The theoretical threshold is about…

15 %

23 %

50 %

65 %

5. A neobank's customer lifetime value (LTV) is calculated as…

CAC × churn

ARPU × contribution margin ÷ monthly churn

ARPU × number of customers

total revenue ÷ CAC

6. Monthly churn rises from 2% to 4%. LTV…

rises by 2%

does not change

doubles

is halved

7. Why does a European neobank earn little on each debit-card payment?

- Because interchange is capped at 0.2% in the EU
- Because Visa keeps the whole fee
- Because the merchant pays nothing
- Because the ECB taxes every payment

8. N26 (2023) earned more revenue per customer than Revolut (2024), and lost money. The main reason:

- its customers paid fewer fees
- interest rates were negative in 2023
- much higher costs per customer, for lack of scale
- the pound is worth more than the euro

SECTION 3.6 · BLOCK 3

🎤 TALK · 20 MIN

Crowdfunding, peer-to-peer lending, buy now pay later

Instructor's talk, or self-study reading.

FOUR FORMS OF CROWDFUNDING

What the person who puts in money gets back

- **Donation**: nothing in return (a school to rebuild, a cause).
- **Reward**: a product or a perk, often the first batch — it is a *pre-sale* (Kickstarter, in Switzerland wemakeit). An "all or nothing" campaign: if the target is not reached, nobody is charged.
- **Lending** (crowdlending): individuals lend to an SME or to another individual, with interest.
- **Equity** (equity crowdfunding): backers become shareholders; they hope for a capital gain on resale, in five to ten years, or nothing if the company fails.

The framework. In the EU, Regulation 2020/1503 (**ECSP**) gives lending and equity platforms an EU-wide licence, for offers of up to **€5 million** per project over twelve months, with a standard key information sheet and a knowledge test for non-sophisticated investors. In Switzerland, professionally accepting funds from more than 20 people normally requires a banking licence; platforms use the exceptions in the Banking Ordinance (funds on a **settlement account** passed on within 60 days) or the fintech licence (Day 5).

THE MECHANISM · A PEER-TO-PEER LOAN

Six steps, from application to repayment

▸ AN SME BORROWS CHF 300,000 FROM 600 INDIVIDUALS

1 · The application. The SME uploads its accounts and its project to the platform: CHF 300,000 over three years for a machine.

2 · The analysis. The platform (not the lenders) analyses the file, assigns a **risk grade** (A to E) and a rate, for example 6.5% for a grade B.

3 · The raise. The project is published; 600 individuals lend CHF 100 to 5,000 each. If the amount is not reached in time, the money is returned.

4 · The payout. The funds pass through an account of the platform or a partner bank — in Switzerland, a *settlement account* that must be emptied within 60 days — then are paid to the SME, minus the platform's commission (2 to 4% of the amount, paid by the borrower).

5 · The repayments. Each month the SME repays; the platform distributes to the 600 lenders, minus its servicing fees (often ~1% a year, paid by the lenders).

6 · Default. If the SME stops paying, the platform chases and collects on the lenders' behalf. The lenders bear the loss; the platform has already collected its commission — hence the incentive problem: it earns on volume, not on quality.

✓ All 6 steps are shown

Expected return ≈ (1 – PD) × rate – PD × LGD – platform fees

The advertised rate is not the return. A loan at 18% with an 18% PD and a 70% LGD yields, after 1% fees, about 1.2% — less than a loan at 9%. And one default in a portfolio of two loans is a disaster; in two hundred, a statistic.

SPLIT PAYMENT · BUY NOW, PAY LATER

"Pay in 4, interest-free": who pays, and is it credit?

- **The mechanism.** At checkout, the provider (Klarna, Afterpay, PayPal Pay in 4, in Switzerland also Powerpay or invoices payable in 30 days) runs a check in a few seconds, pays the merchant at once, and is repaid by the customer in several instalments, often every two weeks.
- **Who pays.** The **merchant**: a fee of a few percent, higher than a card, which it accepts because split payment raises basket size and conversion. The customer pays nothing if they pay on time — and **late fees** otherwise.
- **The status.** In the EU, the old directive excluded interest-free credit repayable within three months; the new one (2023/2225) brings it in from **20 November 2026**, with a creditworthiness assessment. In the UK, the FCA has regulated it since **15 July 2026**. In Switzerland, the KKG/LCC excludes credit repayable within **three months at most**; the

old “four instalments within twelve months” exception disappeared with the 2016 revision.

CHECK · 3 MIN

Which form of crowdfunding?

DONATION · REWARD · LENDING · EQUITY

400 people pay CHF 75 each to receive a bag from the first batch, made in Dakar

Donation Reward Lending Equity

An SME borrows CHF 300,000 at 4.5% over three years from 600 individuals

Donation Reward Lending Equity

120 investors together buy 8% of the shares of a young fintech

Donation Reward Lending Equity

A collection to rebuild a neighbourhood school after a flood

Donation Reward Lending Equity

Backers receive a T-shirt and their name on the project’s website

Donation Reward Lending Equity

One individual lends CHF 2,000 to another, through a platform, to buy a used car

Donation Reward Lending Equity

0 of 6 sorted

SECTION 3.7 · BLOCK 3

🛠 WORKSHOP · 55 MIN

A campaign, a loan portfolio, a debate

A · 10 MIN · THE CAMPAIGN

What is left of a successful campaign

A bag designer in Dakar runs a reward campaign: **400 backers** give **CHF 75** on average. The platform takes **5%**, payment processing **3%**; making the promised bags costs **30%** of the amount raised; shipping costs **CHF 5** per backer.

How much does the designer keep, in CHF, once the platform, payment processing, making the promised bags and shipping are paid?

… CHF Check

B · 20 MIN · THE PEER-TO-PEER LENDING SIMULATOR

Lend CHF 10,000: choose the risk grade, then the number of loans

One-year loans, repaid in one go. A platform grades borrowers from A (safe) to E (risky). If a borrower defaults, you recover 30% (LGD 70%). Platform fees: 1% a year. Each click runs **2,000 simulated years**; the chart shows the distribution of your return.

Risk grade

C · rate 9 % · PD 6 %

Number of loans (CHF 10,000 split equally)

10

GRADE	ADVERTISED RATE	PD	EXPECTED RETURN
A	4 %	1.5 %	to calculate ↓
B	6 %	3 %	to calculate ↓
C	9 %	6 %	to calculate ↓
D	13 %	11 %	to calculate ↓
E	18 %	18 %	to calculate ↓

Simulate 2,000 years →

Expected return on a **grade E** loan (rate 18%, PD 18%, LGD 70%, fees 1%), in %?

… % Check

Same calculation for **grade C** (rate 9%, PD 6%)?

… % Check

✍️ **Your best combination (grade, number of loans) and why. What does diversification change — to the average return, and to the risk of losing?**

The simulation log appears above; summarise it.

C · 25 MIN · STRUCTURED DEBATE

“Is buy now, pay later credit?”

The rules. Two sides drawn by lot: *“yes, and it must be regulated as such”* against *“no, it is a means of payment”*. Five minutes of preparation with the arguments below, two minutes of opening per side, six minutes of exchange, one minute of conclusion per side. Every argument must cite a mechanism or a regulator’s position.

▸ FIRST, SORT THE ARGUMENTS: WHICH SIDE DO THEY SERVE?

The provider pays the merchant at once; the customer repays later

“It is credit” “It is not credit”

No interest is charged to a customer who pays on time

“It is credit” “It is not credit”

A CHF 5 late fee on a CHF 25 instalment, two weeks late

“It is credit” “It is not credit”

It is the merchant, not the customer, who pays the fee

“It is credit” “It is not credit”

A customer can open plans with several providers without any of them seeing the others

“It is credit” “It is not credit”

The amounts are small and the term short: six weeks

“It is credit” “It is not credit”

EU Directive 2023/2225 brings deferred payment into consumer credit from 20 November 2026

“It is credit” “It is not credit”

In the UK, since July 2026 the FCA has required an affordability check before every plan

“It is credit” “It is not credit”

0 of 8 sorted

A CHF 5 late fee on a CHF 25 instalment paid 14 days late: what **annual cost** (simple), in %?

… % Check

Preparation 4:56 Pause ↺

One side’s opening 1:56 Pause ↺

⏱ Exchange **5:56** Pause ↺

✍️ **Your side, the other side’s strongest argument, and your own position after the debate.**

e.g. “Yes” side. Their best argument: the merchant pays… My position: …

DEBRIEF · 15 MIN

The jury (the instructor, or a team drawn by lot) names the side that made the best use of the regulators’ positions. Then a question for the project: could your fintech offer buy now, pay later, and under which licence?

SECTION 3.8 · BLOCK 4

🎤 TALK · 15 MIN

Digital credit in Africa: effects and abuses

Instructor's talk, or self-study reading.

THE FOUNDING CASE · M-SHWARI, KENYA, 2012

A savings account and a loan in the M-Pesa menu

- **The product.** Launched by Commercial Bank of Africa and Safaricom: a savings account and a 30-day loan, from 100 shillings, opened from the M-Pesa menu, with no branch and no paper. The price: a **facilitation fee of 7.5%** of the amount, not an interest rate.
- **The score.** No credit file for most customers: the credit limit is computed from the M-Pesa history (tenure, deposits, transfers, savings on M-Shwari). The first limit is small; it grows with on-time repayments.
- **The measured effect.** Bharadwaj, Jack and Suri (*Journal of Development Economics*, 2021) compare customers just above and just below the eligibility threshold: a third of those eligible borrow; households with access to the loan are **6.3 points** less likely to forgo essential spending after a shock (illness, crop loss). But neither savings nor productive assets rise: digital credit helps absorb a blow, it does not lift people out of poverty.

IN WEST AFRICA

Credit and savings through the mobile wallet

- **Baobab** (a microfinance group, formerly MicroCred) launched **M-Baobab** in Senegal in 2022 with Orange Finances Mobiles: fully digital credit and savings, from the Orange Money app or the USSD code #144#. Baobab+ built a score from the payment history of solar kits paid for by mobile money.
- **The WAEMU framework.** An e-money institution cannot lend (Day 2): the credit is extended by a partner bank or microfinance institution, under the usury rate (14% for banks, 24% for microfinance institutions). That constraint is what sets the West African market apart from Kenya.

THE ABUSES

What went wrong, and what regulators did

- **The hidden price.** A “per loan” fee over 30 days looks modest; annualised, it often exceeds 90%.
- **Over-indebtedness and the blacklist.** In Kenya, borrowers were reported en masse to credit bureaus for defaults of a few hundred shillings, which then shut them out of bank credit.
- **Debt shaming.** Lending apps read the phone’s contact list and wrote to the late borrower’s contacts.
- **The response.** Kenya put digital lenders under central-bank supervision (Central Bank Act amended in 2021, **Digital Credit Providers Regulations** of March 2022): licence required, access to the contact list banned, threats and publication of data banned, no reporting to the credit bureau below 1,000 shillings. In 2025 the Kenyan press still reported harassment by licensed lenders.
- **The opaque score.** A refused customer does not know why, and so can neither challenge it nor improve.

CHECK · 4 MIN

Positive effect, no effect, or abuse?

▸ WHAT RESEARCH AND REGULATORS HAVE DOCUMENTED

Households with access to M-Shwari forgo essential spending less often after a shock

Positive effect | No effect | Abuse

Access to M-Shwari and households’ savings or productive assets

Positive effect | No effect | Abuse

Apps wrote to the phone contacts of a borrower who was late

Positive effect | No effect | Abuse

Borrowers reported to credit bureaus for a few hundred unpaid shillings

Positive effect | No effect | Abuse

Access to M-Pesa and poverty in Kenya (Day 2)

Positive effect | No effect | Abuse

A credit refusal given with no reason at all, by a model nobody can explain

Positive effect | No effect | Abuse

0 of 6 sorted

M-Shwari: a 7.5% fee for a 30-day loan. **Simple annual cost**, in %?

… % Check

And if the borrower takes a new loan as soon as the previous one is repaid, all year long: **compound annual cost**, in %?

… % Check

SECTION 3.9 · BLOCK 4

PROJECT AND PEER REVIEW · 75 MIN

Your app's pension-savings plan

35 MIN · IN TEAMS · THE PRODUCT SHEET

Design the plan, its fees, and what makes it trustworthy

Both teams' project: **an app that lets people in Senegal, Congo or a neighbouring country save for their retirement** by paying small amounts every month until 65, **invested in Swiss blue-chip equity funds**. The app and the company must be transparent and earn trust. Set the simulator with *your* assumptions, then fill in the sheet. Each member fills in their notebook — the same version for the whole team.

WHAT A SMALL MONTHLY PAYMENT BECOMES AT 65 — AND WHAT THE FEES TAKE

Payment per month

10000

Customer's currency

FCFA

Age at the first payment

30

Fund's annual return before fees (%) — an assumption

5

Fund fees, % a year

0.2

Your app's fee, % of assets a year

0.8

Transfer and exchange, % of each payment

1

Swiss franc against the customer's currency, % a year

0

4,200,000 FCFA
PAID IN FROM AGE 30 TO 65

8,846,806 FCFA
CAPITAL AT 65, AFTER ALL FEES

11,129,790 FCFA
THE SAME PLAN WITH NO FEES AT ALL

20.5 %
SHARE OF THE CAPITAL TAKEN BY FEES

6,192,764 FCFA
IF SHARES LOSE 30% IN THE YEAR YOU TURN 65

Payments at the start of each month, return compounded monthly, annual fees deducted from the return. The return is an assumption, not a promise: Swiss large caps also lost more than a third in 2008 and about a quarter within weeks in 2020. The Swiss franc gained value against the euro, and so against the FCFA,

over twenty years (about 1.60 francs to the euro in 2007, 0.93 in 2025), but nothing guarantees what comes next. These are the figures the app must show the customer.

1. The plan: for whom, how much a month and in which currency, through which channel they pay in (Wave, Orange Money, transfer…), into which fund exactly, all the fees (with the share of capital computed above), how the money is paid out at 65 — and what happens if the customer has to withdraw earlier.

e.g. For women traders aged 25 to 45 in Dakar; from 5,000 FCFA a month via Orange Money; an index fund of Swiss large caps (which one, with its fees); app fee …% a year, i.e. …% of the capital at 65; at 65, paid out as …; early withdrawal: …

2. Transparency and trust: who holds the money and the fund units (custodian bank, segregated account), what the customer sees in the app each month (balance, fees in francs, performance), how they can check it without taking your word for it, and the sentence the app shows them the day the fund has lost 20%.

e.g. The units are held in each customer's name at a Swiss custodian bank, separate from the company's accounts; monthly statement: paid in, fees in FCFA, value…; the customer can…; message after a fall: "…"

3. The safeguards: against misappropriation of the funds, against mis-selling (promising a return, pushing a customer who needs their money), against currency risk, against payments being abandoned. One measure for each.

e.g. No return promised, only scenarios…; an external auditor…; a missed payment costs nothing and loses nothing…; …

25 MIN · PEER REVIEW

Another team reads your sheet; you read theirs

Swap screens between the two teams. Read the other team's three fields and answer the five questions below *for them*; tell them your answers out loud. Then write down what your own team received.

THE REVIEW GRID · THE TEAM YOU ARE REVIEWING

The customer is precise: who saves, how much a month, from what age, and why they would do it rather than the tontine.

Yes | To be clarified | No

Fees are given in francs (FCFA, CDF…), not only in %, and their share of the capital at 65 is calculated.

Yes | To be checked | No

We know who holds the money and the fund units, and the customer can check it themselves.

Yes | To be clarified | No

The customer is warned, with a figure, that one year can lose 30% — and knows what happens then.

Yes | Partly | No

At least one real safeguard against misappropriation of funds or mis-selling.

Yes | Too weak | No

0 of 5

Reading and grid **9:56** Pause ↺

4. The criticism received from the other team, and what you change in your product.

e.g. The other team: our fees take 28% of the capital at 65… → app fee cut to 0.5% a year.

BEFORE DAY 4 · FRIDAY 30 OCTOBER

- Day 4 is about blockchain, crypto-assets and decentralised finance. You will create a wallet on a **test** blockchain: nothing to buy, nothing to install in advance; an up-to-date Chrome, Firefox or Brave browser is enough.

- For the project: note whether any part of your app (proof that the units exist, the customer's statement, the transfer from Africa) *could* run on a blockchain. Day 4 will make you say whether it really should.

SECTION 3.10 · BLOCK 4

WORKSHOP · 60 MIN

Build your blockchain

A · 45 MIN · THE CLASS BLOCKCHAIN

Everyone in the room is a node

The pension plan of section 3.9 promises the customer that their units exist and that nobody will rewrite their statement. Before discussing it on Day 4, build a ledger nobody keeps alone: your browser keeps its own copy, checks every block and every payment, and refuses what is false. The site only carries the messages. Seven rounds: keys and signature, hash and block, mining, payments, the chain, cheating, the project.

Open the class blockchain ↗

▸ WHAT THE WORKSHOP RECORDED ON THIS DEVICE

- [ ] Key pair made (round 1)
- [ ] Message signed and verified (round 1)
- [ ] A nonce found by hand (round 2)
- [ ] A block mined on the class chain (round 3)
- [ ] A payment signed and sent (round 4)
- [ ] My copy altered — and refused by the others (round 6 A)
- [ ] A double spend attempted (round 6 B)
- [ ] My copy repaired

The workshop page opens in its own tab; this list updates when you come back here.

B · 15 MIN · FOR THE PROJECT

Which entry of your app deserves this ledger?

What you just built guarantees three things: a signed entry cannot be forged on the customer's behalf; a past entry cannot be rewritten without everyone seeing it; no single

keeper can alter the ledger. It does not guarantee that the units exist at the custodian — the chain proves the entry, not the real world.

✍️ **One entry of your app that would gain from a shared ledger (which one, and who the nodes would be); one that does not need it (and why an ordinary database is enough); and what the ledger will never prove.**

e.g. Each customer's unit statement: nodes = our company, the Swiss custodian, an auditor, the regulator…

MISE EN COMMUN · 10 MIN

On the room screen: the nodes, the longest chain, the balances. Who won the most blocks, and why? What happened to the payment spent twice? And when the instructor altered the site's copy, what did your nodes do?